# Introduction to Linear Programming

## BUS 104: Decision Analysis and Management Science

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# Agenda: Linear Programming Applications

**Welcome!** In this presentation, we will explore the power of Linear Programming (LP) through a series of foundational business and operational problems.

We will cover how to formulate and solve:

1. **Product Mix Problem:** Maximizing manufacturing profit (Par, Inc.)
2. **Blending Problems (I & II):** Mixing ingredients to meet specifications at lowest cost (e.g., Coffee Blending)
3. **Work Scheduling Problem:** Assigning labor shifts to meet demand
4. **Transportation Problem:** Minimizing shipping costs from origins to destinations
5. **Assignment Problem:** Optimal 1-to-1 resource allocation (e.g., Project Staffing)
6. **Shortest Path Problem:** Finding the most efficient route through a network

# Our Approach

For each problem category, we will follow a rigorous 3-step methodology:

- **Step 1: The Business Problem.** Understanding the scenario, the physical constraints, and the managerial objective.
- **Step 2: Mathematical Formulation.** Translating English into Algebra (defining Decision Variables, Objective Function, and Constraints).
- **Step 3: Python Implementation.** Using the `PuLP` library to computationally solve the model and extract business insights.

Let's begin with our first category: The Product Mix Problem.

# Product Mix Example: Par, Inc.

**Goal:** Maximize weekly profit by deciding how many standard and deluxe golf bags to produce.

**The Data:**

| **Department** | **Standard Bag** | **Deluxe Bag** | **Hours Available** |
|---|---|---|---|
| Cutting & Dyeing | 7/10 hr | 1 hr | 630 hrs |
| Sewing | 1/2 hr | 5/6 hr | 600 hrs |
| Finishing | 1 hr | 2/3 hr | 708 hrs |
| Inspection & Pack. | 0.1 hr | 1/4 hr | 135 hrs |
| **Profit/Bag** | **$10** | **$9** | |

- *Assumption:* Demand is unlimited; we can sell everything we make.
- **Question:** Formulate a mathematical model to find the optimal production plan.

# Par, Inc. Golf Bag Production Problem

**Decision variables.**

Let $x_s$ = number of standard golf bags produced per week,
and $x_d$ = number of deluxe golf bags produced per week.

Both $x_s, x_d \geq 0$.

# Par, Inc. Golf Bag Production Problem

**Decision variables.**
Let $x_s$ = number of standard golf bags produced per week,
and $x_d$ = number of deluxe golf bags produced per week.

Both $x_s, x_d \geq 0$.

**Objective function.**
Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d.$$

**Subject to capacity constraints:**

$$\frac{7}{10}x_s + 1x_d \leq 630 \qquad \text{(Cutting \& Dyeing)}$$

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$$\frac{7}{10}x_s + 1x_d \leq 630 \qquad \text{(Cutting \& Dyeing)}$$
$$\frac{1}{2}x_s + \frac{5}{6}x_d \leq 600 \qquad \text{(Sewing)}$$

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**Decision variables.**
Let $x_s$ = number of standard golf bags produced per week,
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Both $x_s, x_d \geq 0$.

**Objective function.**
Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d.$$

**Subject to capacity constraints:**

$$\begin{aligned} \tfrac{7}{10}x_s + 1x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\ \tfrac{1}{2}x_s + \tfrac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\ 1x_s + \tfrac{2}{3}x_d &\leq 708 && \text{(Finishing)} \end{aligned}$$

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**Decision variables.**
Let $x_s$ = number of standard golf bags produced per week,
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**Objective function.**
Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d.$$

**Subject to capacity constraints:**

$$\begin{aligned}
\tfrac{7}{10}x_s + 1x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\
\tfrac{1}{2}x_s + \tfrac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\
1x_s + \tfrac{2}{3}x_d &\leq 708 && \text{(Finishing)} \\
0.1x_s + \tfrac{1}{4}x_d &\leq 135 && \text{(Inspection \& Packaging)} \\
x_s, x_d &\geq 0.
\end{aligned}$$

# Par, Inc. Golf Bag Production Solution (PuLP)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Par_Inc_Golf_Bags", pl.LpMaximize)

# Define variables directly into the model
x_s = prob.add_variable("Standard_Bags", lowBound=0)
x_d = prob.add_variable("Deluxe_Bags", lowBound=0)

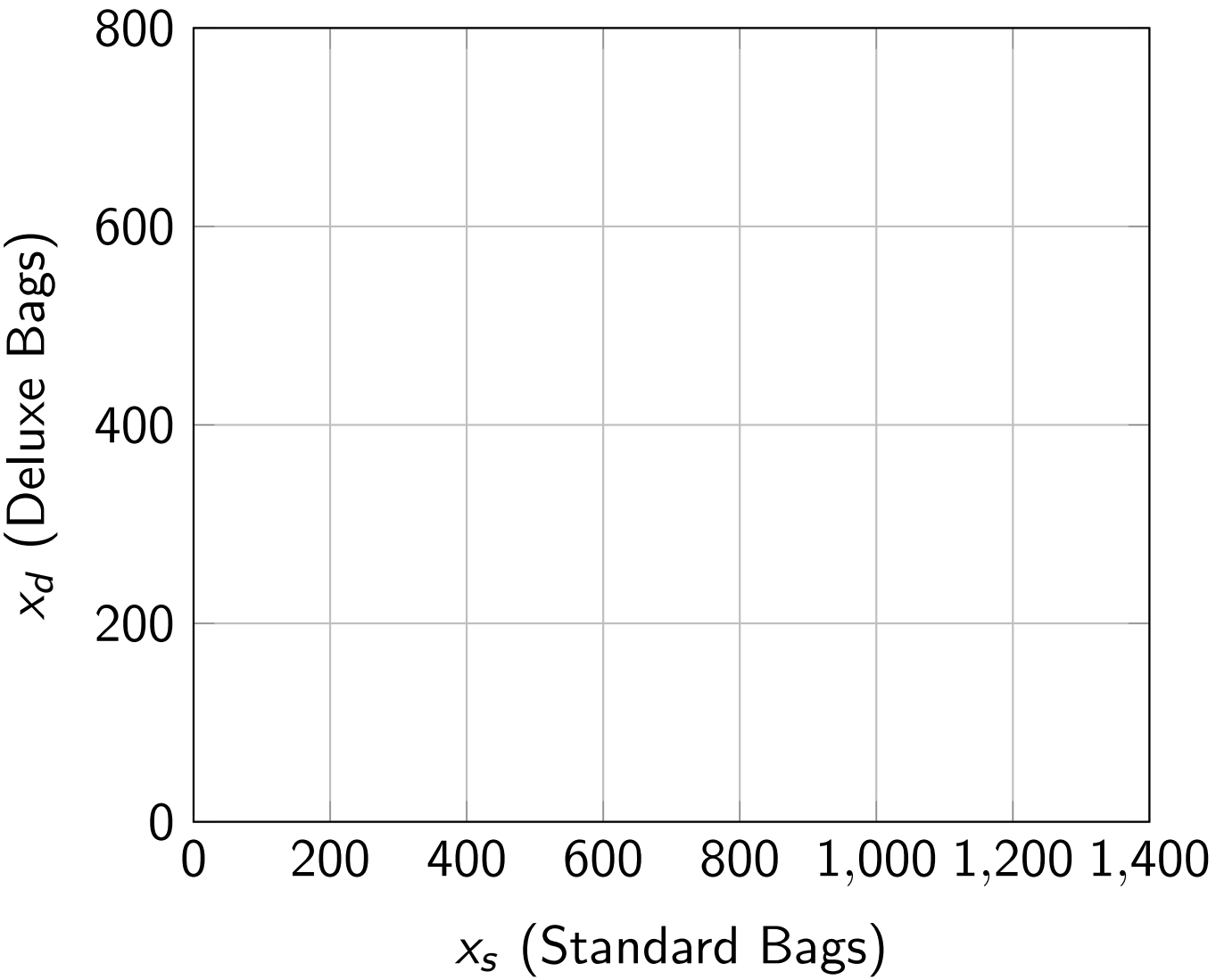
# Objective function
prob += 10 * x_s + 9 * x_d, "Maximize_Profit"

# Constraints
prob += 0.7 * x_s + 1 * x_d <= 630, "Cutting_Dyeing"
prob += 0.5 * x_s + 5/6 * x_d <= 600, "Sewing"
prob += 1 * x_s + 2/3 * x_d <= 708, "Finishing"
prob += 0.1 * x_s + 0.25 * x_d <= 135, "Inspection_Packaging"

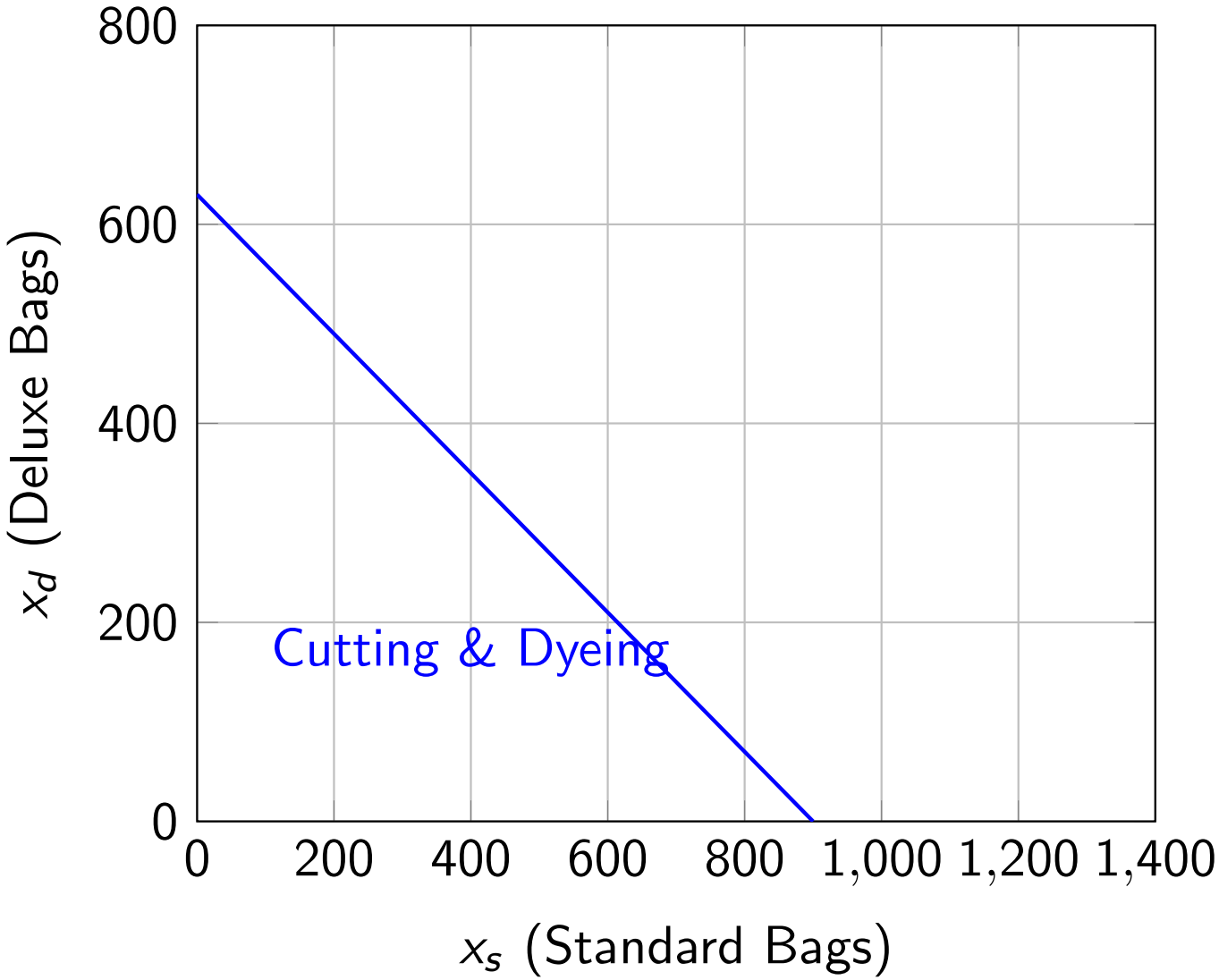
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Standard Bags = {x_s.varValue}")
print(f"Deluxe Bags = {x_d.varValue}")
print(f"Maximum Profit = ${pl.value(prob.objective)}")
```

# Graphical Solution – Step 1: Set up the Axes



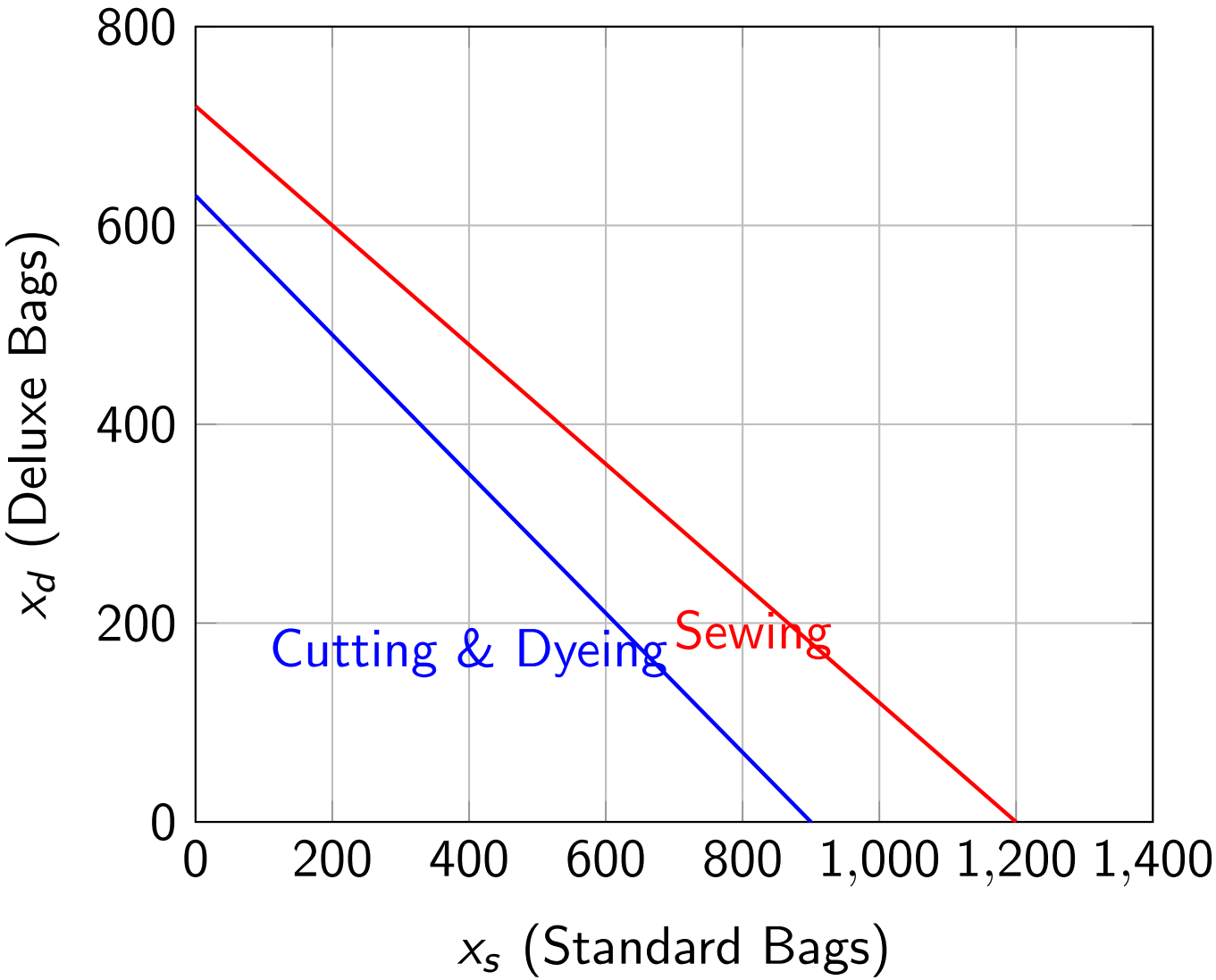
# Graphical Solution – Step 2: Add C&D Constraint




$0.7x_s + x_d \leq 630$

Feasible region is below the line.

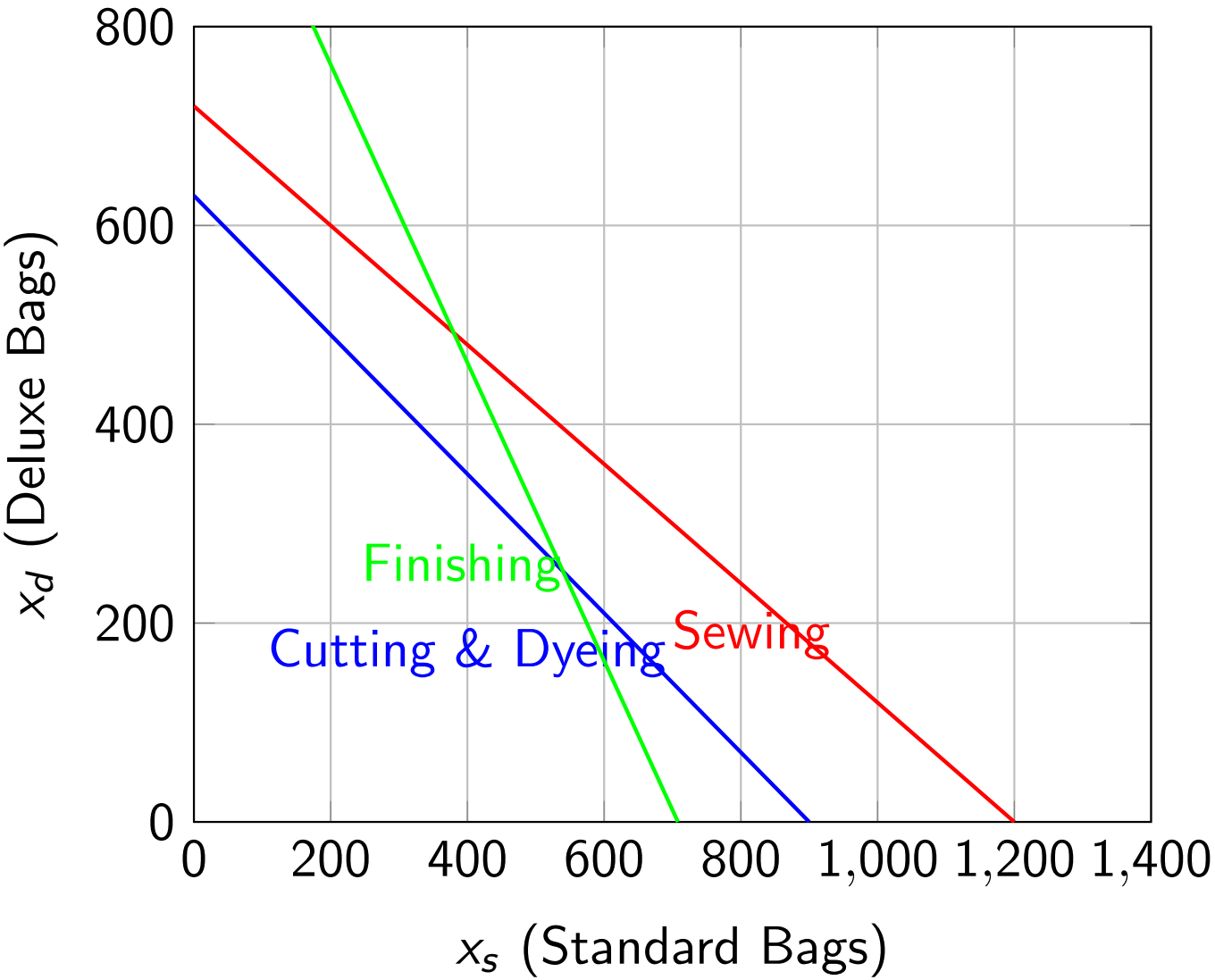
# Graphical Solution – Step 3: Add Sewing Constraint




$0.5x_s + \frac{5}{6}x_d \leq 600$

Feasible region is below the line.

# Graphical Solution – Step 4: Add Finishing Constraint

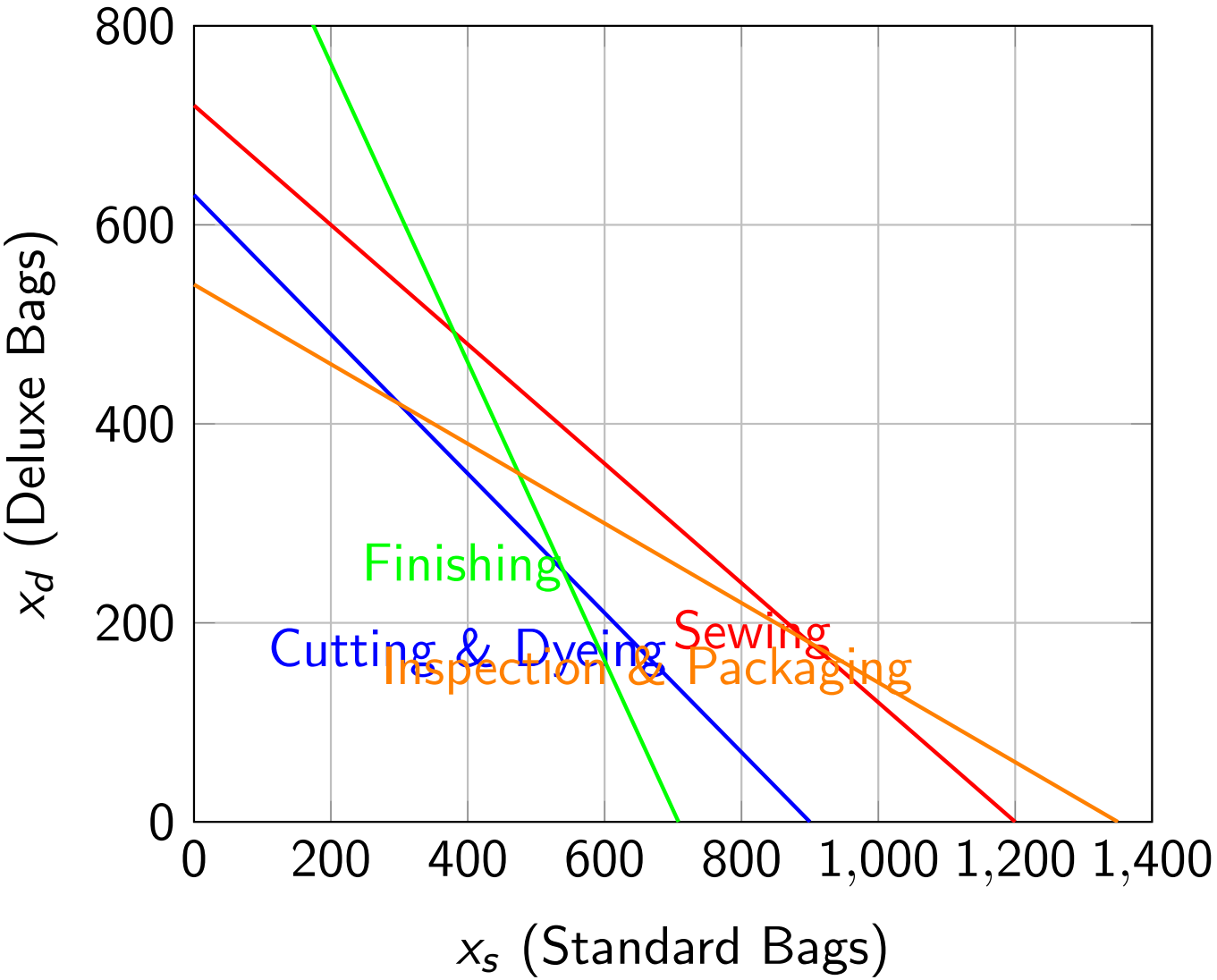



$x_s + \frac{2}{3}x_d \leq 708$

Feasible region is below the line.

# Graphical Solution – Step 5: Add Inspection Constraint

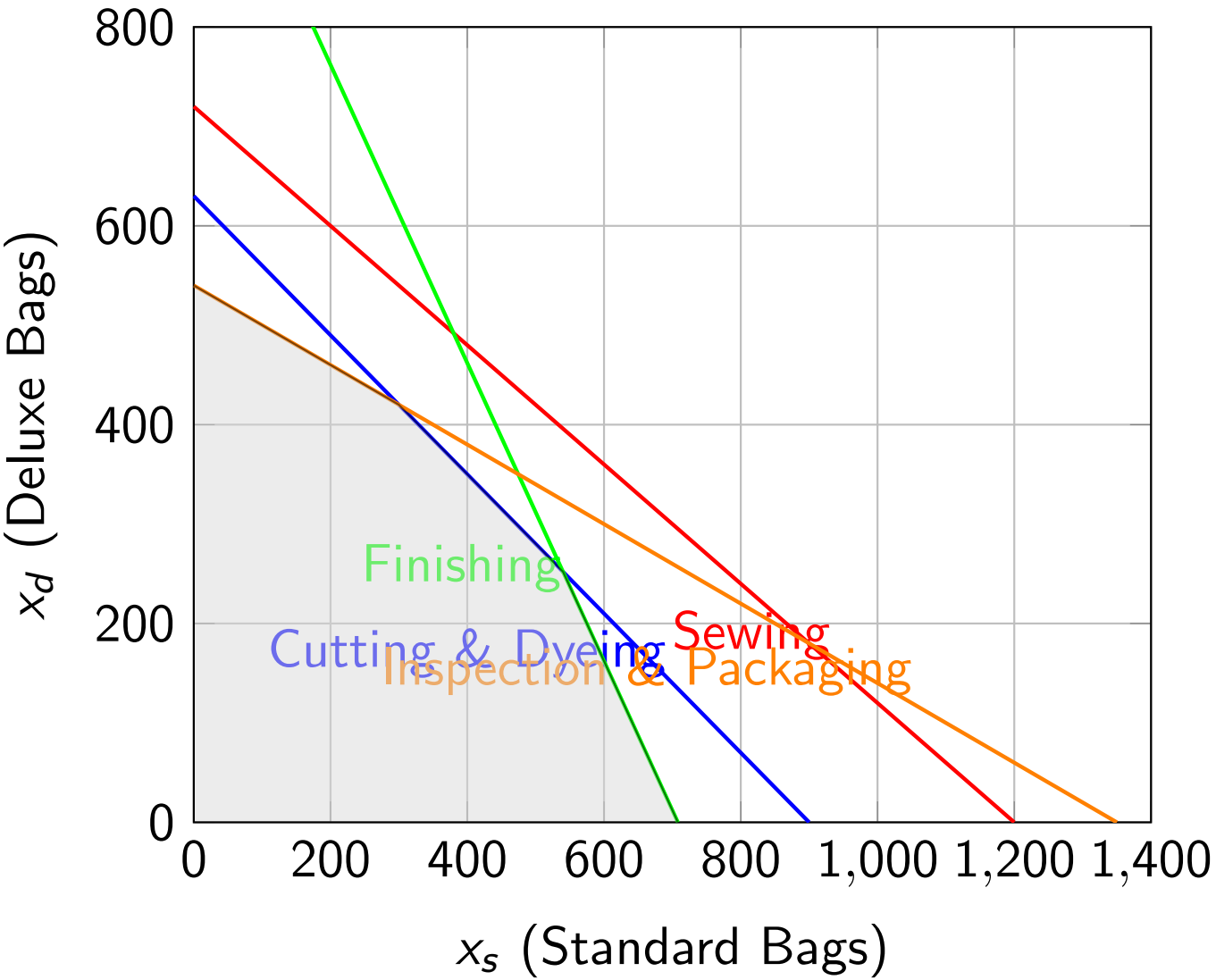
$0.1x_s + 0.25x_d \leq 135$




Feasible region is below the line.

# Graphical Solution – Step 6: Feasible Region

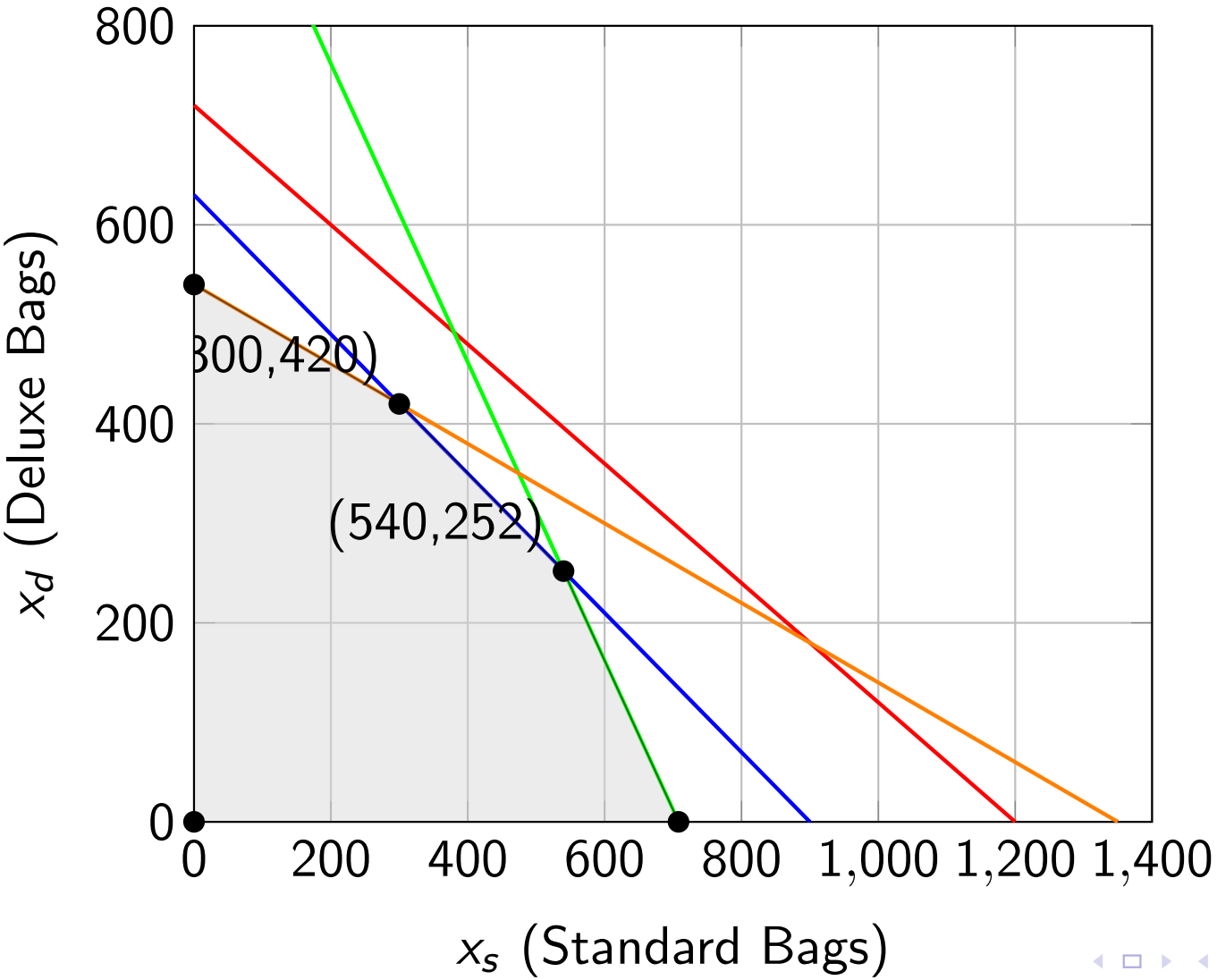
The feasible region is the area below all constraint lines and above the




axes.

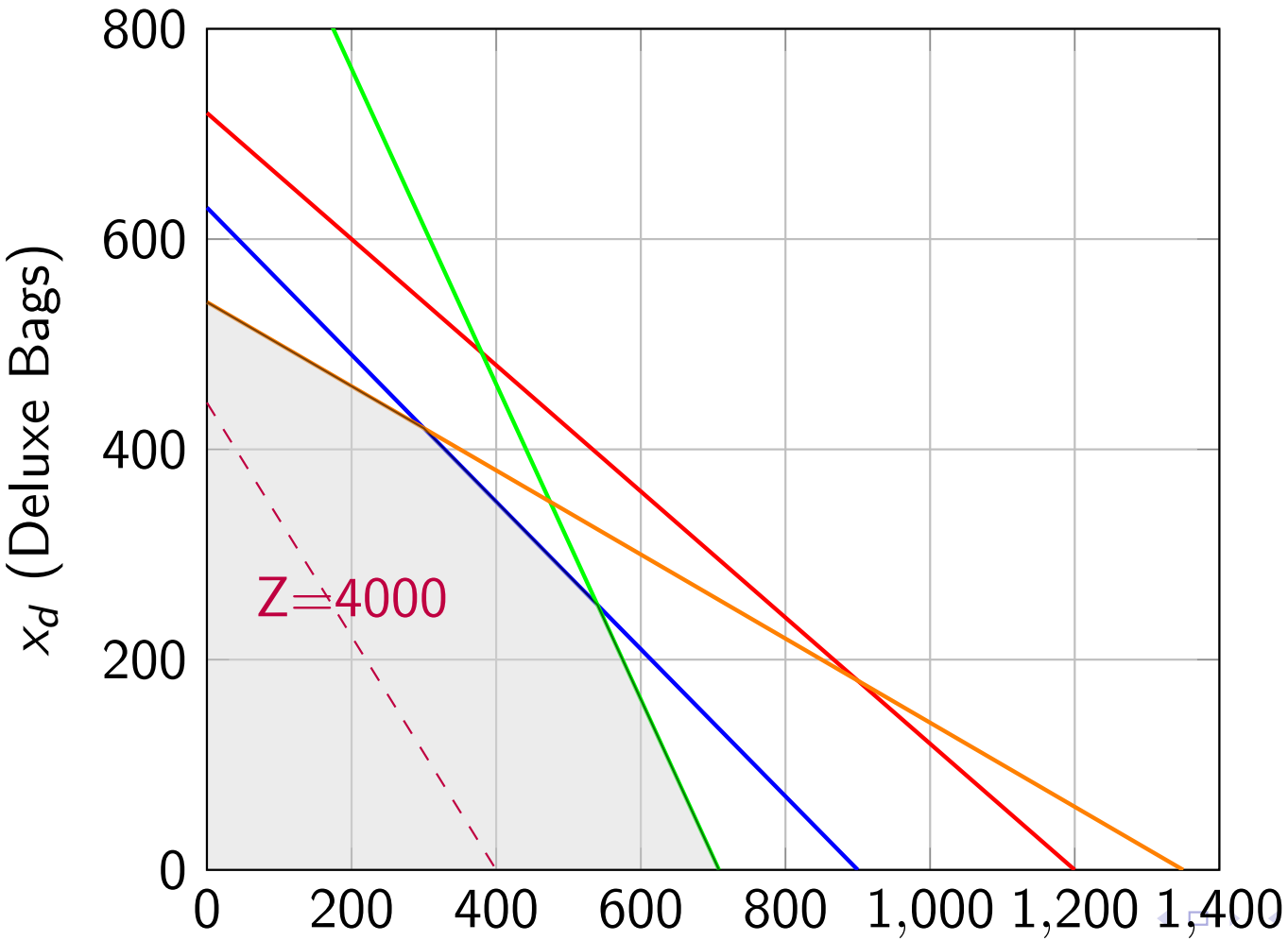
# Graphical Solution – Step 7: Corner Points

Identify the corner points of the feasible region: - (0, 0) - (0, 540) - (300, 420) - (540, 252) - (708, 0)



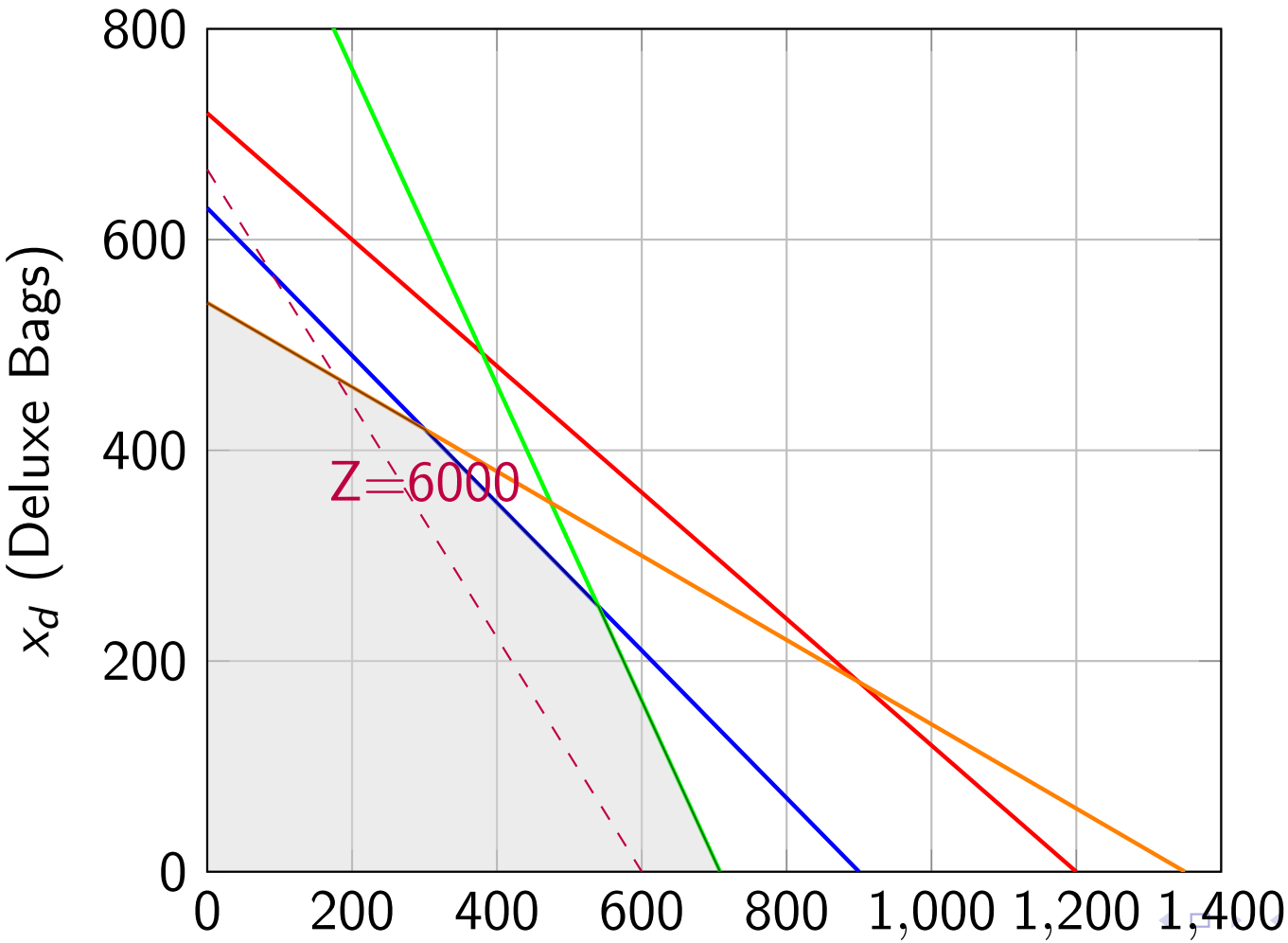
# Graphical Solution – Step 8: Objective Function at Low Profit

Objective: $Z = 10x_s + 9x_d = 4000$ Line:
$x_d = \frac{4000}{9} - \frac{10}{9}x_s \approx 444.44 - 1.111x_s$



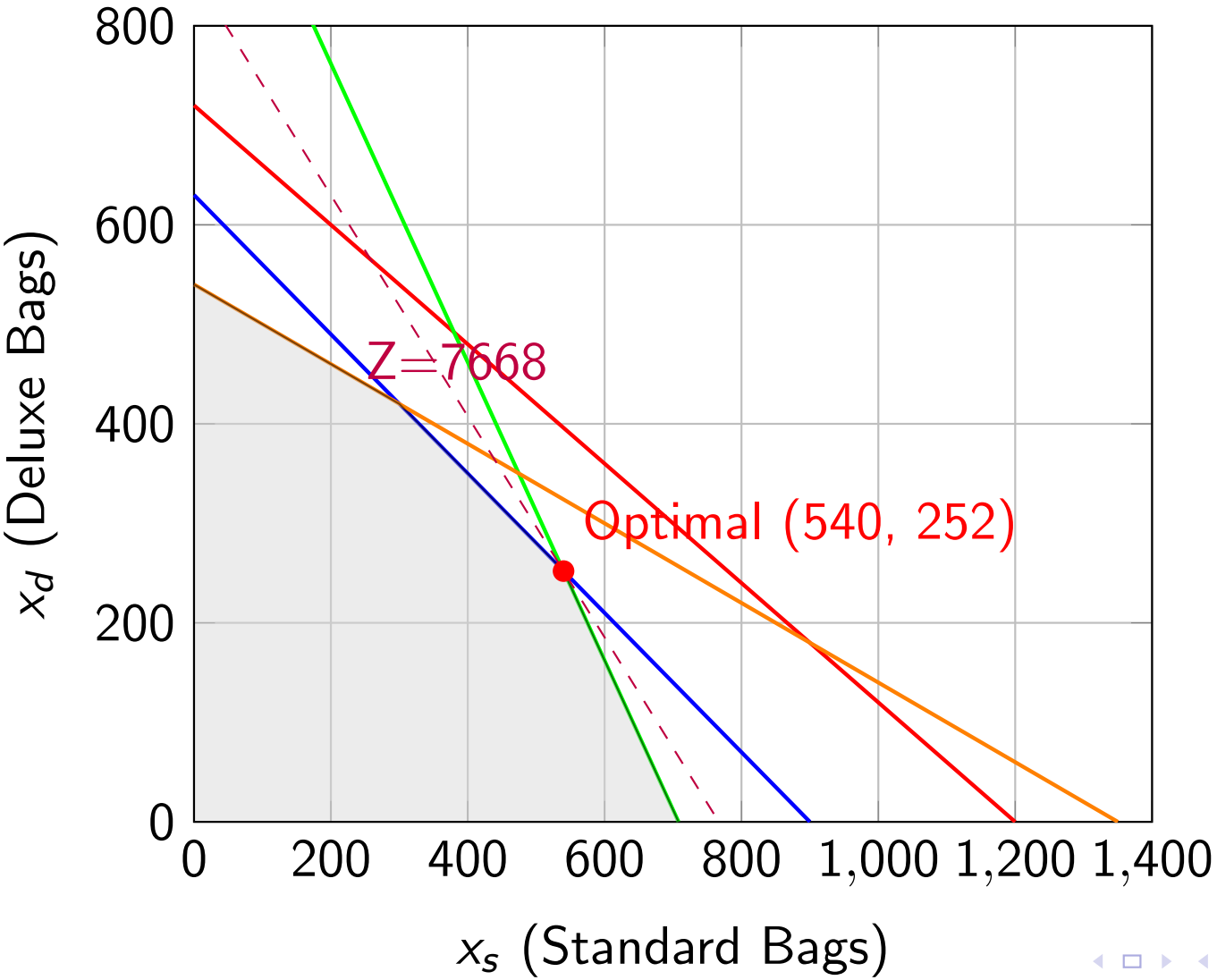
# Graphical Solution – Step 9: Objective Function at Higher Profit

Objective: $Z = 10x_s + 9x_d = 6000$ Line:
$x_d = \frac{6000}{9} - \frac{10}{9}x_s \approx 666.67 - 1.111x_s$



# Graphical Solution – Step 10: Optimal Objective Function

Objective: $Z = 10x_s + 9x_d = 7668$ Line: $x_d = \frac{7668}{9} - \frac{10}{9}x_s \approx 852 - 1.111x_s$
The line touches the feasible region at the optimal point (540, 252).



# Blending Example: MediBlend

**Goal:** Find the cheapest way to produce today's batch of MediBlend by mixing four chemicals.

**The Rules & Constraints:**

- **Batch Size:** Must produce *at least* 1,000 lbs total.
- **Ingredient Minimums:** The final mix must contain at least 8% of A, 4% of B, and 2% of C (by weight).
- **Supply Constraint:** Must use *at least* 100 lbs of Chemical 2.

**The Data:**

| Chemical | Cost/lb | Ingredient A | Ingredient B | Ingredient C |
|---|---|---|---|---|
| Chemical 1 | \$8 | 0.03 | 0.02 | 0.01 |
| Chemical 2 | \$10 | 0.06 | 0.04 | 0.01 |
| Chemical 3 | \$11 | 0.10 | 0.03 | 0.04 |
| Chemical 4 | \$14 | 0.12 | 0.09 | 0.04 |

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

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Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\min_{x_1,x_2,x_3,x_4} \quad 8x_1 + 10x_2 + 11x_3 + 14x_4$$

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \end{aligned}$$

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Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\ & 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \end{aligned}$$

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned} \min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\ & 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\ & 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \end{aligned}$$

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4)
\end{aligned}$$

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4) \\
& x_2 \geq 100
\end{aligned}$$

# Blending Example: LP Formulation

Let $x_1, x_2, x_3, x_4$ denote the pounds of chemicals (1,2,3,4) used to produce today's batch. The objective is to minimize total cost subject to meeting the minimum batch size, the ingredient percentage requirements, the minimum usage of chemical 2, and nonnegativity:

$$\begin{aligned}
\min_{x_1,x_2,x_3,x_4} \quad & 8x_1 + 10x_2 + 11x_3 + 14x_4 \\
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 \\
& 0.03x_1 + 0.06x_2 + 0.10x_3 + 0.12x_4 \geq 0.08(x_1 + x_2 + x_3 + x_4) \\
& 0.02x_1 + 0.04x_2 + 0.03x_3 + 0.09x_4 \geq 0.04(x_1 + x_2 + x_3 + x_4) \\
& 0.01x_1 + 0.01x_2 + 0.04x_3 + 0.04x_4 \geq 0.02(x_1 + x_2 + x_3 + x_4) \\
& x_2 \geq 100 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}$$

# Blending Example: Classical LP Formulation

In classical Linear Programming (and for many solvers), we rearrange our equations so that all variables are on the left side of the inequality, and all constants are on the right side.

By factoring out the variables from the right side of our ingredient constraints, our model simplifies to:

$$\min_{x_1,x_2,x_3,x_4} \quad 8x_1 + 10x_2 + 11x_3 + 14x_4$$

$$\begin{aligned}
\text{s.t.} \quad & x_1 + x_2 + x_3 + x_4 \geq 1000 && \text{(Batch Size)} \\
& -0.05x_1 - 0.02x_2 + 0.02x_3 + 0.04x_4 \geq 0 && \text{(Ingredient A)} \\
& -0.02x_1 - 0.01x_3 + 0.05x_4 \geq 0 && \text{(Ingredient B)} \\
& -0.01x_1 - 0.01x_2 + 0.02x_3 + 0.02x_4 \geq 0 && \text{(Ingredient C)} \\
& x_2 \geq 100 && \text{(Chemical 2 Min)} \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}$$

# Blending Example Solution (PuLP)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("MediBlend_Blending", pl.LpMinimize)

# Define variables directly into the model
x1 = prob.add_variable("Chemical_1", lowBound=0)
x2 = prob.add_variable("Chemical_2", lowBound=0)
x3 = prob.add_variable("Chemical_3", lowBound=0)
x4 = prob.add_variable("Chemical_4", lowBound=0)

# Objective function
prob += 8 * x1 + 10 * x2 + 11 * x3 + 14 * x4, "Minimize_Cost"

# Constraints
prob += x1 + x2 + x3 + x4 >= 1000, "Batch_Size"
prob += 0.03 * x1 + 0.06 * x2 + 0.10 * x3 + 0.12 * x4 >= 0.08 * (x1 + x2 + x3 + x4), "
    Ingredient_A"
prob += 0.02 * x1 + 0.04 * x2 + 0.03 * x3 + 0.09 * x4 >= 0.04 * (x1 + x2 + x3 + x4), "
    Ingredient_B"
prob += 0.01 * x1 + 0.01 * x2 + 0.04 * x3 + 0.04 * x4 >= 0.02 * (x1 + x2 + x3 + x4), "
    Ingredient_C"
prob += x2 >= 100, "Chemical_2_Min"

# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Chemical 1 = {x1.varValue} lbs")
print(f"Chemical 2 = {x2.varValue} lbs")
print(f"Chemical 3 = {x3.varValue} lbs")
print(f"Chemical 4 = {x4.varValue} lbs")
print(f"Minimum Cost = ${pl.value(prob.objective)}")
```

# Work Scheduling Example

A post office requires different numbers of full-time employees on different days of the week. The number of full-time employees needed on each day is as follows:

| Day | Number of Employees |
|---|---|
| Monday | 7 |
| Tuesday | 3 |
| Wednesday | 5 |
| Thursday | 9 |
| Friday | 4 |
| Saturday | 6 |
| Sunday | 1 |

Table: Weekly Employee Requirements

Union rules state that each full-time employee must work five consecutive days and then receive two days off. For example, an employee who works Monday to Friday must be off Saturday and Sunday. The post office wants to meet its daily requirements using only full-time employees. Formulate an LP that the post office can use to minimize the number of full-time employees that must be hired.

# Work Scheduling: Shift Coverage Grid

To formulate the constraints, it helps to visualize which days each shift covers. Employees work 5 consecutive days and then get 2 days off.

| **Shift Starts** | **Mon** | **Tue** | **Wed** | **Thu** | **Fri** | **Sat** | **Sun** |
|---|---|---|---|---|---|---|---|
| Monday ($x_{\text{Mon}}$) | W | W | W | W | W | Off | Off |
| Tuesday ($x_{\text{Tue}}$) | Off | W | W | W | W | W | Off |
| Wednesday ($x_{\text{Wed}}$) | Off | Off | W | W | W | W | W |
| Thursday ($x_{\text{Thu}}$) | W | Off | Off | W | W | W | W |
| Friday ($x_{\text{Fri}}$) | W | W | Off | Off | W | W | W |
| Saturday ($x_{\text{Sat}}$) | W | W | W | Off | Off | W | W |
| Sunday ($x_{\text{Sun}}$) | W | W | W | W | Off | Off | W |
| **Required** | **7** | **3** | **5** | **9** | **4** | **6** | **1** |

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$\min \quad x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}}$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$\begin{aligned} \min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} \\ \text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 \qquad \text{(Mon)} \end{aligned}$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)} \\
& x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} \geq 1 & \text{(Sun)}
\end{aligned}
$$

# Work Scheduling Example: LP Formulation

Let $(x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}})$ denote the number of full-time employees whose 5-day workweek begins on the indicated day (each employee works 5 consecutive days, then has 2 consecutive days off). The goal is to minimize the total number of employees started this week subject to daily staffing requirements and nonnegativity:

$$
\begin{aligned}
\min \quad & x_{\text{Mon}} + x_{\text{Tue}} + x_{\text{Wed}} + x_{\text{Thu}} + x_{\text{Fri}} + x_{\text{Sat}} + x_{\text{Sun}} & \\
\text{s.t.} \quad & x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} \geq 7 & \text{(Mon)} \\
& x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} \geq 3 & \text{(Tue)} \\
& x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} + x_{\text{Sat}} \geq 5 & \text{(Wed)} \\
& x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} + x_{\text{Sun}} \geq 9 & \text{(Thu)} \\
& x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} + x_{\text{Mon}} \geq 4 & \text{(Fri)} \\
& x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} + x_{\text{Tue}} \geq 6 & \text{(Sat)} \\
& x_{\text{Sun}} + x_{\text{Sat}} + x_{\text{Fri}} + x_{\text{Thu}} + x_{\text{Wed}} \geq 1 & \text{(Sun)} \\
& x_{\text{Mon}}, x_{\text{Tue}}, x_{\text{Wed}}, x_{\text{Thu}}, x_{\text{Fri}}, x_{\text{Sat}}, x_{\text{Sun}} \geq 0. &
\end{aligned}
$$

# Work Scheduling Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Post_Office_Scheduling", pl.LpMinimize)

# Define variables directly into the model
x_mon = prob.add_variable("x_Mon", lowBound=0, cat='Integer')
x_tue = prob.add_variable("x_Tue", lowBound=0, cat='Integer')
x_wed = prob.add_variable("x_Wed", lowBound=0, cat='Integer')
x_thu = prob.add_variable("x_Thu", lowBound=0, cat='Integer')
x_fri = prob.add_variable("x_Fri", lowBound=0, cat='Integer')
x_sat = prob.add_variable("x_Sat", lowBound=0, cat='Integer')
x_sun = prob.add_variable("x_Sun", lowBound=0, cat='Integer')

# Objective function
prob += x_mon + x_tue + x_wed + x_thu + x_fri + x_sat + x_sun, "Total_Employees"

# Constraints
prob += x_mon + x_sun + x_sat + x_fri + x_thu >= 7, "Monday"
prob += x_tue + x_mon + x_sun + x_sat + x_fri >= 3, "Tuesday"
prob += x_wed + x_tue + x_mon + x_sun + x_sat >= 5, "Wednesday"
prob += x_thu + x_wed + x_tue + x_mon + x_sun >= 9, "Thursday"
prob += x_fri + x_thu + x_wed + x_tue + x_mon >= 4, "Friday"
prob += x_sat + x_fri + x_thu + x_wed + x_tue >= 6, "Saturday"
prob += x_sun + x_sat + x_fri + x_thu + x_wed >= 1, "Sunday"
```

# Work Scheduling Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Employees starting Mon = {x_mon.varValue}")
print(f"Employees starting Tue = {x_tue.varValue}")
print(f"Employees starting Wed = {x_wed.varValue}")
print(f"Employees starting Thu = {x_thu.varValue}")
print(f"Employees starting Fri = {x_fri.varValue}")
print(f"Employees starting Sat = {x_sat.varValue}")
print(f"Employees starting Sun = {x_sun.varValue}")
print(f"Minimum Employees = {pl.value(prob.objective)}")
```

# Blending Example II

The Java Junkie Coffee Shop blends coffee on the premises for its customers. It sells three basic blends in 1-pound bags: Special, Mountain Dark, and Mill Regular. It uses two different types of beans to produce the blends: Brazilian and mild. The shop used the following blend recipe requirements.

| BLEND | MIX REQUIREMENTS | SELLING PRICE ($/LB) |
|---|---|---|
| Special | At least 40% Brazilian No more 30% mild | 6.50 |
| Dark | At least 60% Brazilian No more than 10% mild | 5.25 |
| Regular | No more than 60% mild At least 30% Brazilian | 3.75 |

The cost of Brazilian coffee beans are $2.00 per pound and the cost of mild is $1.25 per pound. Every week the shop can buy up-to 210 pounds of Brazilian coffee beans, and 300 pounds of mild coffee beans. The shop wants to know the amount of each blend it should prepare each week and how much of each type of coffee beans to include in each blend in order to maximize profits. Assume the weekly demand to all three blends always exceeds the store's supply.

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\max \quad 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR})$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\ & - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR})\end{aligned}$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}),
\end{aligned}$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}),
\end{aligned}$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& \quad - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}),
\end{aligned}$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210,
\end{aligned}$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$
\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& \quad - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210, \\
& \text{Mild availability: } x_{mS} + x_{mD} + x_{mR} \leq 300,
\end{aligned}
$$

# Blending Example II: LP with Double-Indexed Variables

Let $(x_{bS}, x_{bD}, x_{bR})$ denote pounds of Brazilian beans used in Special, Dark, and Regular, and $(x_{mS}, x_{mD}, x_{mR})$ denote pounds of mild beans used in those blends. The objective maximizes profit (revenue minus bean costs), with recipe, availability, and nonnegativity constraints:

$$
\begin{aligned}
\max \quad & 6.50(x_{bS} + x_{mS}) + 5.25(x_{bD} + x_{mD}) + 3.75(x_{bR} + x_{mR}) \\
& - 2.00(x_{bS} + x_{bD} + x_{bR}) - 1.25(x_{mS} + x_{mD} + x_{mR}) \\
\text{s.t.} \quad & \text{Special: } x_{bS} \geq 0.40(x_{bS} + x_{mS}), \quad x_{mS} \leq 0.30(x_{bS} + x_{mS}), \\
& \text{Dark: } x_{bD} \geq 0.60(x_{bD} + x_{mD}), \quad x_{mD} \leq 0.10(x_{bD} + x_{mD}), \\
& \text{Regular: } x_{mR} \leq 0.60(x_{bR} + x_{mR}), \quad x_{bR} \geq 0.30(x_{bR} + x_{mR}), \\
& \text{Brazilian availability: } x_{bS} + x_{bD} + x_{bR} \leq 210, \\
& \text{Mild availability: } x_{mS} + x_{mD} + x_{mR} \leq 300, \\
& x_{bS}, x_{bD}, x_{bR}, x_{mS}, x_{mD}, x_{mR} \geq 0.
\end{aligned}
$$

Equivalently, the recipe constraints may be written in purely linear form:

$$
\begin{aligned}
& 0.60x_{bS} - 0.40x_{mS} \geq 0, \quad -0.30x_{bS} + 0.70x_{mS} \leq 0, \\
& 0.40x_{bD} - 0.60x_{mD} \geq 0, \quad -0.10x_{bD} + 0.90x_{mD} \leq 0, \\
& -0.60x_{bR} + 0.40x_{mR} \leq 0, \quad 0.70x_{bR} - 0.30x_{mR} \geq 0.
\end{aligned}
$$

# Coffee Blending Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Java_Junkie_Coffee", pl.LpMaximize)

# Define variables directly into the model
x_bS = prob.add_variable("Brazilian_Special", lowBound=0)
x_bD = prob.add_variable("Brazilian_Dark", lowBound=0)
x_bR = prob.add_variable("Brazilian_Regular", lowBound=0)
x_mS = prob.add_variable("Mild_Special", lowBound=0)
x_mD = prob.add_variable("Mild_Dark", lowBound=0)
x_mR = prob.add_variable("Mild_Regular", lowBound=0)

# Objective function
prob += (6.50 * (x_bS + x_mS) + 5.25 * (x_bD + x_mD) + 3.75 * (x_bR + x_mR)
         - 2.00 * (x_bS + x_bD + x_bR) - 1.25 * (x_mS + x_mD + x_mR)), "Maximize_Profit"

# Constraints
# Special blend constraints
prob += x_bS >= 0.40 * (x_bS + x_mS), "Special_Brazilian_Min"
prob += x_mS <= 0.30 * (x_bS + x_mS), "Special_Mild_Max"

# Dark blend constraints
prob += x_bD >= 0.60 * (x_bD + x_mD), "Dark_Brazilian_Min"
prob += x_mD <= 0.10 * (x_bD + x_mD), "Dark_Mild_Max"

# Regular blend constraints
prob += x_mR <= 0.60 * (x_bR + x_mR), "Regular_Mild_Max"
prob += x_bR >= 0.30 * (x_bR + x_mR), "Regular_Brazilian_Min"

# Availability constraints
prob += x_bS + x_bD + x_bR <= 210, "Brazilian_Availability"
prob += x_mS + x_mD + x_mR <= 300, "Mild_Availability"
```

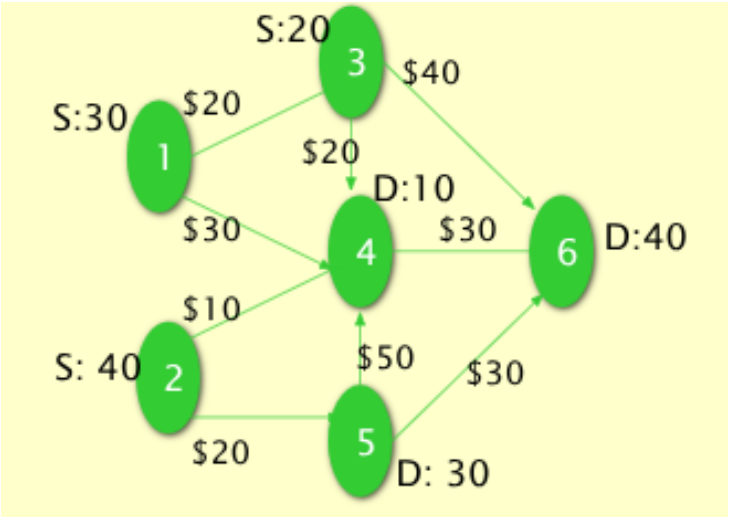
# Coffee Blending Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"Brazilian in Special = {x_bS.varValue} lbs")
print(f"Brazilian in Dark = {x_bD.varValue} lbs")
print(f"Brazilian in Regular = {x_bR.varValue} lbs")
print(f"Mild in Special = {x_mS.varValue} lbs")
print(f"Mild in Dark = {x_mD.varValue} lbs")
print(f"Mild in Regular = {x_mR.varValue} lbs")
print(f"Maximum Profit = ${pl.value(prob.objective)}")
```

# Network Model: Transportation Problem

A furniture manufacturer, Furniture Max, has three warehouses (1, 2, 3) , and three retail stores (4, 5, 6). The following network diagram indicates the shipping routes and shipping costs. The values on the arcs indicate the per unit shipping costs required to transport furniture between the various cities. The supply of furniture at each warehouse is indicated by the number next to nodes 1, 2, and 3. The demand for furniture is indicated by the number next to the nodes 4, 5, 6. Formulate an LP to determine the least costly shipping plan for Furniture Max.



# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

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$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

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**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

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$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

Node 1: $x_{13} + x_{14} \leq 30$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

Node 1: $x_{13} + x_{14} \leq 30$

Node 2: $x_{24} + x_{25} \leq 40$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \; 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

$$\text{Node 6: } x_{36} + x_{56} + x_{46} \geq 40$$

# Transportation Problem

**Decisions** $x_{ij}$ = units shipped from node $i$ to node $j$

**Objective**

$$\min \ 20x_{13} + 30x_{14} + \cdots + 30x_{56}$$

$$\text{Node 1: } x_{13} + x_{14} \leq 30$$

$$\text{Node 2: } x_{24} + x_{25} \leq 40$$

$$\text{Node 3: } x_{36} + x_{34} - x_{13} \leq 20$$

$$\text{Node 4: } x_{14} + x_{24} + x_{34} + x_{54} - x_{46} \geq 10$$

$$\text{Node 5: } x_{25} - x_{54} - x_{56} \geq 30$$

$$\text{Node 6: } x_{36} + x_{56} + x_{46} \geq 40$$

$$x_{ij} \geq 0$$

# Transportation Problem Solution (PuLP) (1/2)

```
import pulp as pl

# Create the problem
prob = pl.LpProblem("Furniture_Max_Transportation", pl.LpMinimize)

# Define variables directly into the model
x13 = prob.add_variable("x13", lowBound=0)
x14 = prob.add_variable("x14", lowBound=0)
x24 = prob.add_variable("x24", lowBound=0)
x25 = prob.add_variable("x25", lowBound=0)
x34 = prob.add_variable("x34", lowBound=0)
x36 = prob.add_variable("x36", lowBound=0)
x46 = prob.add_variable("x46", lowBound=0)
x54 = prob.add_variable("x54", lowBound=0)
x56 = prob.add_variable("x56", lowBound=0)

# Objective function
prob += 20 * x13 + 30 * x14 + 10 * x24 + 20 * x25 + 30 * x34 + 40 * x36 + 30 * x46 + 50 *
      x54 + 30 * x56, "Minimize_Cost"

# Constraints
# Supply constraints
prob += x13 + x14 <= 30, "Node_1_Supply"
prob += x24 + x25 <= 40, "Node_2_Supply"
prob += x34 + x36 - x13 <= 20, "Node_3_Supply"

# Demand constraints
prob += x14 + x24 + x34 + x54 - x46 >= 10, "Node_4_Demand"
prob += x25 - x54 - x56 >= 30, "Node_5_Demand"
prob += x36 + x56 + x46 >= 40, "Node_6_Demand"
```

# Transportation Problem Solution (PuLP) (2/2)

```
# Solve the problem
res = prob.solve()

# Print results
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"x13 = {x13.varValue}")
print(f"x14 = {x14.varValue}")
print(f"x24 = {x24.varValue}")
print(f"x25 = {x25.varValue}")
print(f"x34 = {x34.varValue}")
print(f"x36 = {x36.varValue}")
print(f"x46 = {x46.varValue}")
print(f"x54 = {x54.varValue}")
print(f"x56 = {x56.varValue}")
print(f"Minimum Cost = ${pl.value(prob.objective)}")
```

## Consulting Staffing Problem

A firm must assign 4 Senior Project Leaders to 4 Client Projects (1-to-1 matching).

| Leader | P1 | P2 | P3 | P4 |
|---|---|---|---|---|
| Adams | 38 | 75 | 44 | 27 |
| Baker | 34 | 76 | 43 | 25 |
| Clark | 41 | 71 | 41 | 26 |
| Davis | 33 | 80 | 45 | 30 |

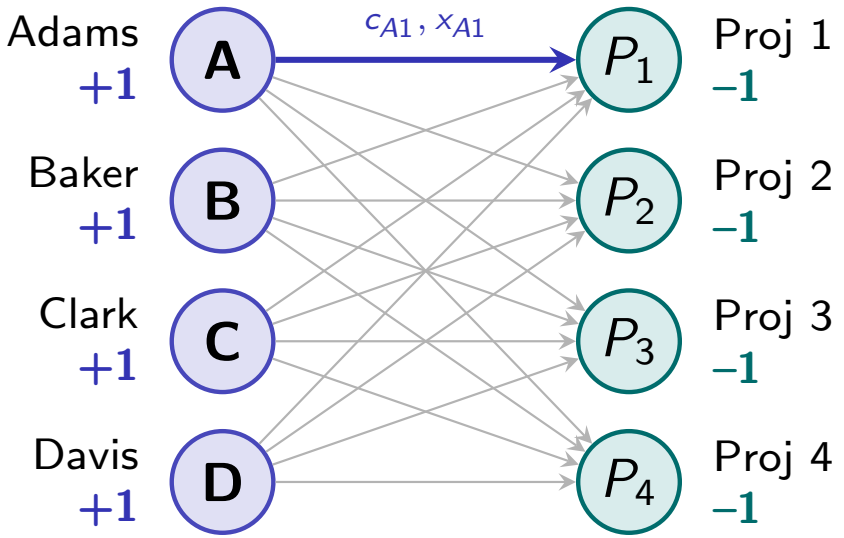
*Values represent completion times in hours.*

## Model Characteristics

- **1-to-1 Matching:** Bipartite assignment between $n$ agents and $n$ tasks.
- **Objective:** Minimize total consulting hours across all projects.
- **Combinatorial Scale:** $4! = 24$ permutations. For $n = 10$, $10! \approx 3.63 \times 10^6$.
- **LP Advantage:** Solves in milliseconds without brute-force search!

# The Assignment Problem as a Network Flow

## Bipartite Graph Formulation




**Decision Arcs:** $x_{ij} \in \{0, 1\}$ with unit cost $c_{ij}$ hours

### Network Flow Properties

- **Specialized Transportation Model:** Bipartite network with source nodes $I$ (supply $s_i = 1$) and sink nodes $J$ (demand $d_j = 1$).
- **Balanced Flow Conservation:** $\sum_{i=1}^{4} s_i = \sum_{j=1}^{4} d_j = 4$.
- **Unit Capacity Arcs:** Flow across arc $(i, j)$ is either 0 or 1.
- **Decision Variable:** $x_{ij} = 1$ if leader $i \rightarrow$ project $j$; 0 otherwise.

# Assignment Problem: Mathematical Formulation

## Binary Integer Program vs. Continuous Linear Program

### Optimization Model

$$\min \; Z = \sum_{i=1}^{4} \sum_{j=1}^{4} c_{ij} x_{ij}$$

**Subject to:**

- $\sum_{j=1}^{4} x_{ij} = 1 \quad \forall i$ (1 proj / leader)
- $\sum_{i=1}^{4} x_{ij} = 1 \quad \forall j$ (1 leader / proj)
- $x_{ij} \geq 0 \quad \forall i, j$ (Non-negativity)

### Total Unimodularity (TUM)

- **TUM Incidence Matrix:** Bipartite node-arc matrix is totally unimodular.
- **Guaranteed Integrality:** Integer RHS ($\pm 1$) guarantees binary solutions ($x_{ij}^* \in \{0, 1\}$).
- **Solves as LP:** Continuous Simplex guarantees integer solutions instantly!

# Assignment Problem in Python: PuLP (1/2)

## Problem Definition & Parameter Setup

```
import pulp as pl

# 1. Initialize problem
prob = pl.LpProblem("Project_Assignment", pl.LpMinimize)

# 2. Sets and completion hours matrix
leaders = ['Adams', 'Baker', 'Clark', 'Davis']
projects = ['P1', 'P2', 'P3', 'P4']
costs = {
    'Adams': {'P1': 38, 'P2': 75, 'P3': 44, 'P4': 27},
    'Baker': {'P1': 34, 'P2': 76, 'P3': 43, 'P4': 25},
    'Clark': {'P1': 41, 'P2': 71, 'P3': 41, 'P4': 26},
    'Davis': {'P1': 33, 'P2': 80, 'P3': 45, 'P4': 30}
}

# 3. Decision variables: x[i, j] in {0, 1}
x = pl.LpVariable.dicts("Assign", (leaders, projects), cat='Binary'
    )

# 4. Objective: Minimize total hours
prob += pl.lpSum(costs[i][j] * x[i][j] for i in leaders for j in
    projects)
```

### Implementation Mechanics

- `pl.LpVariable.dicts()`: Creates 16 binary variables $x_{ij}$ over leaders $\times$ projects.
- `pl.lpSum()`: Compactly generates the double summation $\sum_i \sum_j c_{ij} x_{ij}$.

# Assignment Problem in Python: PuLP (2/2)

## Constraints, Solution & Optimal Staffing Schedule

```
# Each leader gets 1 project
for i in leaders:
    prob += pl.lpSum(x[i][j] for j
        in projects) == 1

# Each project gets 1 leader
for j in projects:
    prob += pl.lpSum(x[i][j] for i
        in leaders) == 1

# Solve optimization model
res = prob.solve()
print("Status:", res.status.name
    )
print(f"Min Hours = {pl.value(
    prob.objective)}")
```

### Optimal Schedule ($Z^* = 173$ h)

| Leader | Project | Hours |
|---|---|---|
| Adams | Project 3 | 44 h |
| Baker | Project 4 | 25 h |
| Clark | Project 2 | 71 h |
| Davis | Project 1 | 33 h |

### Managerial Insight

- **Global Optimization:** Assigning Baker to P4 (25 h) frees Adams for P3 (44 h).
- **Bottleneck Match:** Clark must lead P2 (71 h); others incur a 4–9 h penalty.

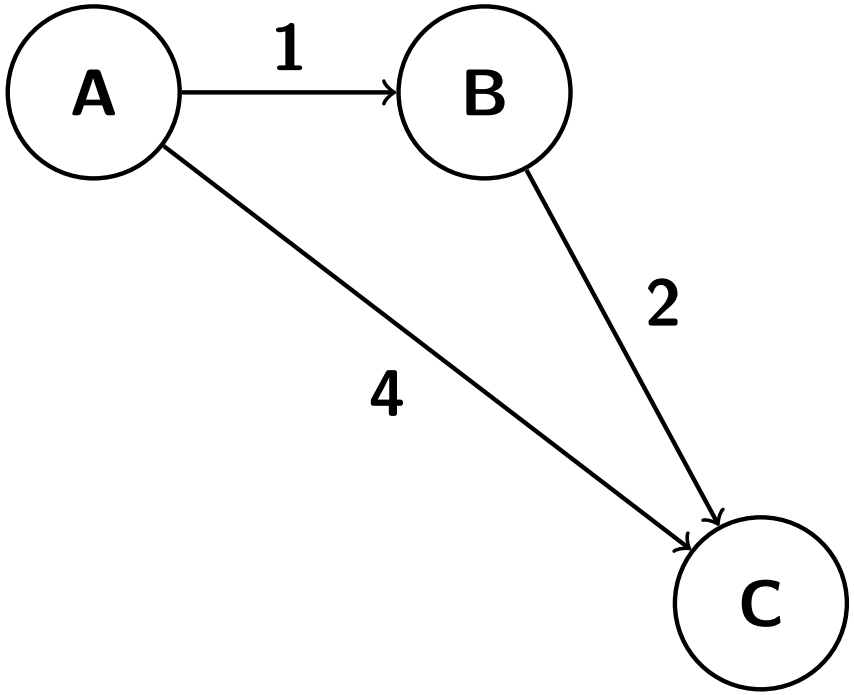
# The Shortest Path Problem

LP and an Heuristic

- Using a toy example

# The Shortest Path Problem: An Example

Find the shortest path from A to C.




**Edges with costs:**

- A to B: **1**
- A to C: **4**
- B to C: **2**

# Problem Statement

- The shortest path problem seeks the path from source (A) to sink (C) with minimal total cost.
- Traditional algorithms: Dijkstra, Bellman-Ford.
- Here, we formulate it as a Linear Program (LP) using network flow.

# LP Formulation: Variables and Objective

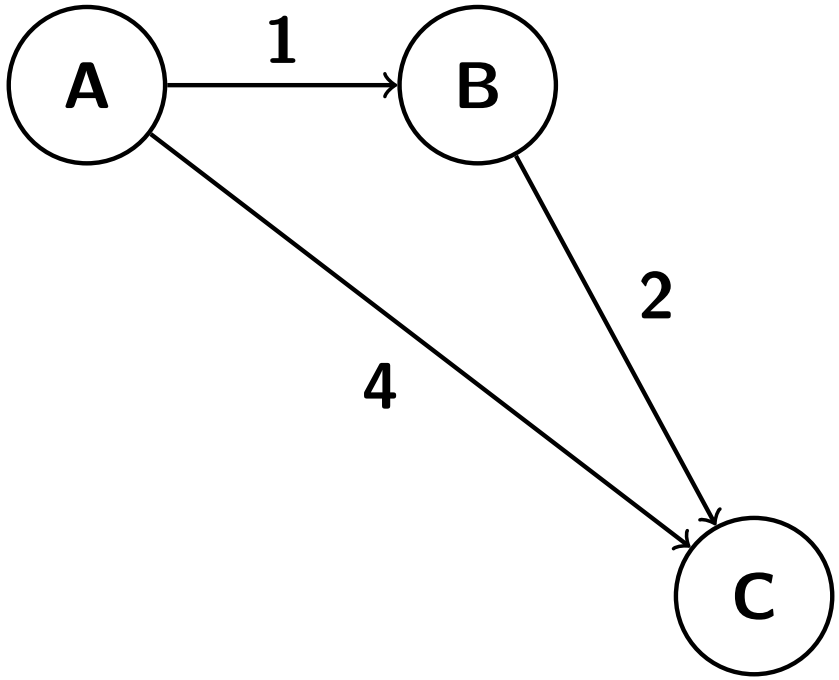
Variables (flow on edges):

- $x_{AB}$: flow from A to B
- $x_{AC}$: flow from A to C
- $x_{BC}$: flow from B to C

Objective: Minimize total cost

$$\min \ 1 \cdot x_{AB} + 4 \cdot x_{AC} + 2 \cdot x_{BC}$$

# LP Formulation: Constraints




Flow conservation (net flow out):

- Source A: net out = 1 $\quad x_{AB} + x_{AC} = 1$
- Intermediate B: net out = 0 $\quad x_{BC} - x_{AB} = 0$
- Sink C: net out = -1 $\quad -(x_{AC} + x_{BC}) = -1$ or $x_{AC} + x_{BC} = 1$

Non-negativity: $x_{AB}, x_{AC}, x_{BC} \geq 0$

# Solving with PuLP

```
import pulp as pl

prob = pl.LpProblem("ShortestPath", pl.LpMinimize)

x_ab = prob.add_variable("x_ab", lowBound=0)
x_ac = prob.add_variable("x_ac", lowBound=0)
x_bc = prob.add_variable("x_bc", lowBound=0)

prob += 1*x_ab + 4*x_ac + 2*x_bc

prob += x_ab + x_ac == 1, "Node_A"
prob += x_bc - x_ab == 0, "Node_B"
prob += x_ac + x_bc == 1, "Node_C"

res = prob.solve()
```

# Optimal Solution

- $x_{AB} = 1$
- $x_{AC} = 0$
- $x_{BC} = 1$
- Total Cost $= 3$

Path: A $\rightarrow$ B $\rightarrow$ C (cost $1 + 2 = 3$)

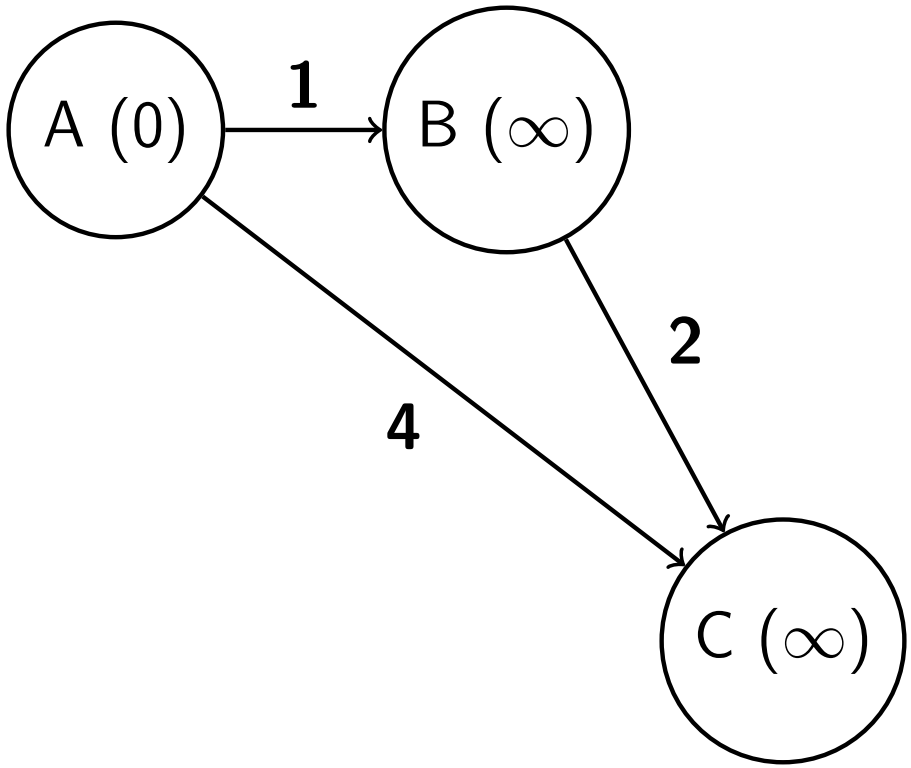
# Conclusion

- The LP formulation models the shortest path as a min-cost flow problem with unit supply/demand.
- Advantages: Handles variants (e.g., capacities); solvable with LP solvers.
- Note: For pure shortest path, integer solutions are guaranteed due to network structure.

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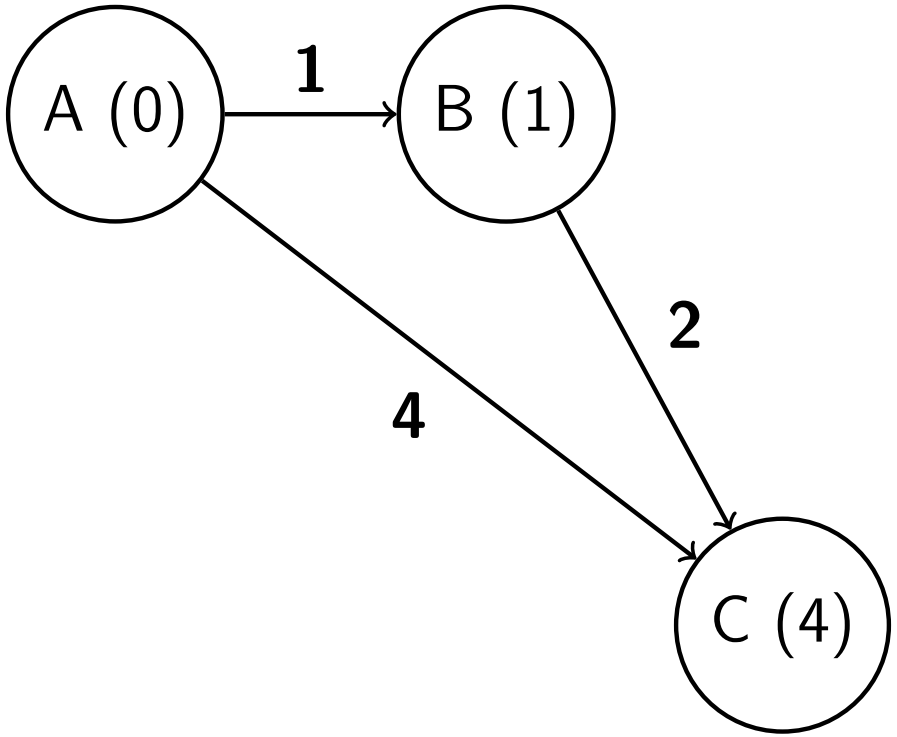
# Dijkstra's Algorithm: Step 1 - Initialization

Find the shortest path from A to C.



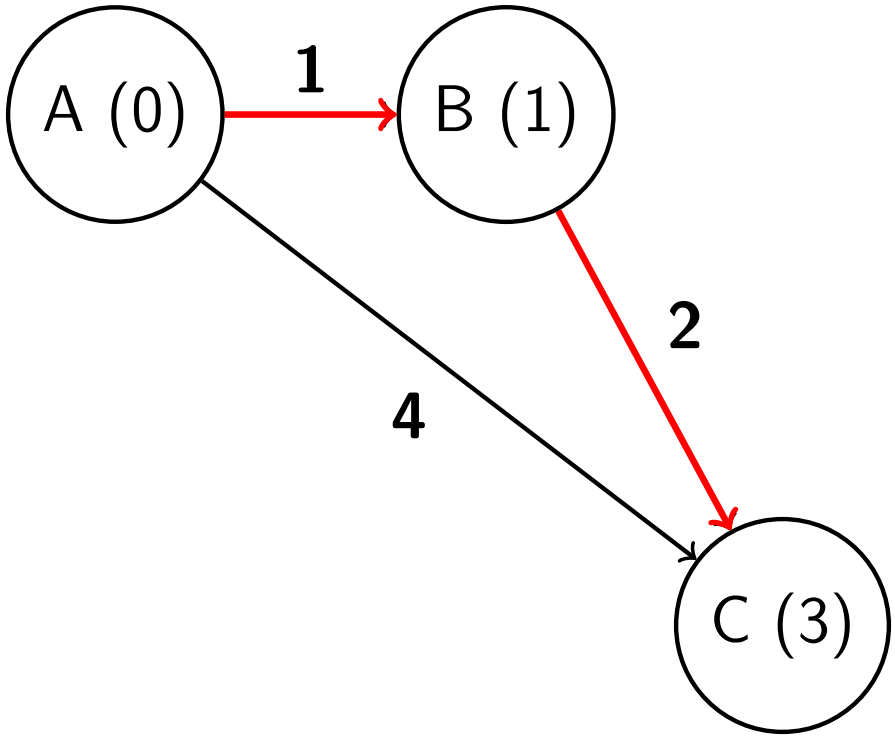

- Start at node A, set distance to 0. - Initialize distances to all other nodes as $\infty$. - Set A as the current node.

# Dijkstra's Algorithm: Step 2 - Update Distances




- Update distances from A: - B: $0 + 1 = 1$ (update from $\infty$) - C: $0 + 4 = 4$ (update from $\infty$) - Mark A as visited. - Move to unvisited node with smallest distance: B (1).

# Dijkstra's Algorithm: Step 3 - Final Path




- Update distances from B: - C: $\min(4, 1+2) = 3$ (update from 4) - Mark B as visited. - Only unvisited node is C (3), mark as visited. - Shortest path: A $\rightarrow$ B $\rightarrow$ C with total cost 3.

# Sensitivity Analysis in Linear Programming

Shadow Prices and Reduced Costs

- Using Par, Inc. Golf Bag Production as Example

# Introduction to Sensitivity Analysis

Sensitivity analysis examines how changes in the LP parameters (coefficients in objective, RHS of constraints, etc.) affect the optimal solution.

## Why important?

- Helps understand the robustness of the solution.
- Provides managerial insights (e.g., value of resources).
- Identifies critical parameters.

## Key concepts:

- Shadow prices (for constraints)
- Reduced costs (for variables)
- Ranges of validity

# Recap of Par, Inc. Problem

**Decision variables:**

Let $x_s$ = number of standard golf bags produced per week,
$x_d$ = number of deluxe golf bags produced per week.
Both $x_s, x_d \geq 0$.

**Objective function:**

Maximize weekly profit:

$$\max \ Z = 10x_s + 9x_d$$

**Subject to capacity constraints:**

$$\begin{aligned}
\frac{7}{10}x_s + x_d &\leq 630 && \text{(Cutting \& Dyeing)} \\
\frac{1}{2}x_s + \frac{5}{6}x_d &\leq 600 && \text{(Sewing)} \\
x_s + \frac{2}{3}x_d &\leq 708 && \text{(Finishing)} \\
0.1x_s + 0.25x_d &\leq 135 && \text{(Inspection \& Packaging)} \\
x_s, x_d &\geq 0
\end{aligned}$$

# Writing the PuLP Code

## Gemini Prompt

Here is an linear program: max $Z = 10x_s + 9x_d$ s.t. $\frac{7}{10}x_s + x_d \leq 630; \frac{1}{2}x_s + \frac{5}{6}x_d; x_s + \frac{2}{3}x_d \leq 708; 0.1x_s + 0.25x_d \leq 135; x_s, x_d \geq 0$. Write me a python code that solves the linear program using PuLP. Also add code that will compute **reduced costs**, **shadow prices**, and **indicate which constraints are loose or tight**.

# PuLP Code with Shadow Prices and Reduced Costs

```
import pulp as pl

# Model
prob = pl.LpProblem("Par_Inc_Golf_Bags", pl.LpMaximize)

# Variables (continuous  0; add cat='Integer' if you ever need integrality)
x_s = prob.add_variable("Standard_Bags", lowBound=0)
x_d = prob.add_variable("Deluxe_Bags",   lowBound=0)

# Objective
prob += 10 * x_s + 9 * x_d, "Maximize_Profit"

# Constraints
prob += 0.7 * x_s + 1.0   * x_d <= 630, "Cutting_Dyeing"
prob += 0.5 * x_s + (5/6) * x_d <= 600, "Sewing"
prob += 1.0 * x_s + (2/3) * x_d <= 708, "Finishing"
prob += 0.1 * x_s + 0.25  * x_d <= 135, "Inspection_Packaging"

# Solve with CBC so reduced costs are available
res = prob.solve()
```

# PuLP Code with Shadow Prices and Reduced Costs

```
# Report primal solution
print("Status:", res.status.name)
print("Optimal Solution:")
print(f"  Standard Bags = {x_s.varValue:.6f}")
print(f"  Deluxe Bags   = {x_d.varValue:.6f}")
print(f"  Maximum Profit = ${pl.value(prob.objective):.6f}")

# Reduced costs for variables (dj):
# For a maximization, a nonbasic variable has dj <= 0.
print("\nReduced Costs (variables.dj):")
for v in prob.variables():
    dj_str = f"{v.dj:.6f}" if v.dj is not None else "None"
    print(f"  {v.name:20s}  dj = {dj_str}")

# Shadow prices (constraint.pi) and slacks (constraint.slack)
# For a <= constraint, slack = RHS - LHS at the optimum.
print("\nShadow Prices and Slacks:")
TOL = 1e-7  # numerical tolerance for binding check
for c in prob.constraints():
    cname = c.name
    dual = c.pi
    slack = c.slack
    # Classify binding if slack ~ 0
    is_binding = abs(slack) <= TOL
    print(f"  {cname:22s}  dual = {dual: .6f}   slack = {slack: .6f}   binding = {
        is_binding}")

# If you also want a compact "which constraints bind" list:
binding_constraints = [name for name, c in prob.constraints.items() if abs(c.slack) <=
    TOL]
print("\nBinding constraints:", binding_constraints if binding_constraints else "None")
```

# Optimal Solution

No need to read the code. Just run it to get this:

## Optimal Production Plan

- **Standard Bags** ($x_s$): 540
- **Deluxe Bags** ($x_d$): 252

## Maximum Profit

$$\$7,668.00$$

## Binding Constraints

These resources are fully utilized (Slack $= 0$) and define the optimal solution.

- **Cutting/Dyeing** ($\mathbf{C_1}$): $0.7x_s + 1.0x_d \leq 630$
- **Finishing** ($\mathbf{C_3}$): $1.0x_s + \frac{2}{3}x_d \leq 708$

# Sensitivity Analysis & Managerial Decision-Making

## Post-Optimality, Dual Prices, and Hurdle Rates

### Part A: Foundations (Par, Inc.)

- **Shadow Prices:** Value of scarce capacity
- **Rule of Slack:** Idle capacity $\Longrightarrow$ \$0 value
- **Reduced Costs:** Hurdle rates for new items
- **Sensitivity Ranges:** Optimality & Feasibility

### Part B: Case Application (Parket)

- **PuLP Duals:** Extracting dual values in Python
- **Material Sourcing:** Valuing extra inventory
- **Supply Shocks:** Absorbing cuts via slack
- **R&D Redesign:** Evaluating a new launch

### Core Managerial Goal

Equip managers to make immediate, profitable decisions as conditions change.

# Beyond the Optimal Solution: Why Sensitivity?

## Fast Decision-Making in Volatile Operating Environments

### The Manager's Dilemma

- Business conditions shift constantly: prices change, machines break, supplies delay.
- Re-solving large enterprise LPs for every minor shift is slow and costly.
- **Goal:** Extract instant marginal insights directly from current optimal solution.

### Two Core Business Questions

1. **Resource Capacity:**
   - *"Max willingness to pay for an extra hour of bottleneck labor?"*

   $\rightarrow$ **Shadow Price (Dual Value)**
2. **Product Portfolio:**
   - *"How profitable must a new product be to justify diverting capacity?"*

   $\rightarrow$ **Reduced Cost**

# Extracting Dual Values in Python with PuLP

## Direct Access to Shadow Prices, Slack, and Reduced Costs

### PuLP Sensitivity Extraction Code

```
# Query constraint shadow prices and slacks
for c in prob.constraints.values():
  print(f"{c.name}:  Shadow Price = ${abs(c.pi):.4f}, Slack =
{abs(c.slack):.2f}")
# Query variable reduced costs
for v in prob.variables():
  print(f"{v.name}:  Reduced Cost = ${v.dj:.4f}")
```

### c.pi: Shadow Price

Marginal profit per $+1$ capacity unit: $\pi_i = \frac{\Delta Z^*}{\Delta b_i}$

### c.slack: Slack

Unused capacity at the optimal plan: $s_i = b_i - \sum_j a_{ij} x_j^*$

### v.dj: Reduced Cost

Profit shortfall vs. resource hurdle rate: $d_j = c_j - z_j$

# Par, Inc.: Departmental Resource Audit

Optimal Production: 540 Standard, 252 Deluxe ($Z^* = \$7,668$)

| Department | Available | Used | Slack | Shadow Price | Status |
|---|---|---|---|---|---|
| Cutting & Dyeing | 630 hrs | 630 hrs | 0 hrs | \$4.375 / hr | **Bottleneck** |
| Sewing | 600 hrs | 480 hrs | 120 hrs | \$0.000 / hr | Surplus |
| Finishing | 708 hrs | 708 hrs | 0 hrs | \$6.938 / hr | **Top Bottleneck** |
| Inspection & Pkg | 135 hrs | 117 hrs | 18 hrs | \$0.000 / hr | Surplus |

## Bottlenecks (Slack = 0)

- Cutting and Finishing are at 100% capacity utilization.
- **Finishing** delivers highest return on expansion (**\$6.94/hr**).

## The Golden Rule of Slack

- Positive slack ($s_i > 0$) always implies Shadow Price = **\$0**.
- Idle hours add no profit: Sewing and Inspection yield \$0.

# Valuing Capacity: Overtime & Investment Decisions

Translating Shadow Prices into Practical Labor Strategy

## Sunk vs. Variable Labor

**Case 1: Sunk / Fixed Wages**

- Base pay budgeted in overhead.
- **Shadow Price = Max Total Wage**.
- *Cutting: Pay up to \$4.38/hr total.*

**Case 2: Variable Base Wages**

- Margin already deducts base pay.
- **Shadow Price = Max OT Premium**.

## Finishing Overtime Example

**Proposal:** $+2$ hrs Finishing ($\pi_3 = \$6.94$).

- $\Delta Z^* = 2 \text{ hrs} \times \$6.9375 = +\mathbf{\$13.88}$
- New Profit = $\$7,668 + \$13.88 = \mathbf{\$7,681.88}$

**Decision Rule:**
Approve if incremental overtime cost is under **\$13.88** total.

# Reduced Cost: The Hurdle Rate for New Products

## Evaluating Whether Unproduced Items Justify Capacity Reallocation

### Reduced Cost ($d_j$): The Scarce Resource Hurdle Rate

The profit premium an unproduced item must achieve to justify taking bottleneck capacity away from existing products:

$$\mathbf{d_j} = \mathbf{c_j} - \sum_{\mathbf{i=1}}^{\mathbf{m}} \mathbf{a_{ij}}\pi_\mathbf{i} = \text{Unit Profit} - \text{Resource Opportunity Cost}$$

### Produced Products ($x_j^* > 0$)

- Clears resource hurdle rate exactly.
- **Reduced Cost:** $\mathbf{d_j} = \$\mathbf{0.00}$.
- Earns its spot in optimal production.

### Unproduced Products ($x_j^* = 0$)

- Fails to clear resource hurdle rate.
- **Reduced Cost:** $\mathbf{d_j} \leq \$\mathbf{0.00}$.
- Forcing 1 unit destroys $|d_j|$ in profit.

# The "Standard Plus" Bag: Pricing Out Resources

Evaluating a Proposed Product Addition for Par, Inc.

## Marketing Proposal: New "Standard Plus" Model

Requires: 0.80 hr Cutting, 0.60 hr Sewing, 0.80 hr Finishing, and 0.15 hr Inspection.

| Department | Usage | Opp. Cost |
|---|---|---|
| Cutting & Dyeing | 0.80 hr | \$3.50 |
| Sewing (Surplus) | 0.60 hr | \$0.00 |
| Finishing | 0.80 hr | \$5.55 |
| Inspection (Surplus) | 0.15 hr | \$0.00 |
| **Total Hurdle Rate:** | | **\$9.05** |

*Surplus hours contribute \$0.00 opportunity cost!*

## Scenario A: Profit = \$10.00

$$d = \$10.00 - \$9.05 = +\mathbf{\$0.95} > \mathbf{0}$$

**Decision: PRODUCE!** Clears hurdle rate; increases company profit.

## Scenario B: Profit = \$8.50

$$d = \$8.50 - \$9.05 = -\mathbf{\$0.55} < \mathbf{0}$$

**Decision: REJECT!** Fails hurdle rate; destroys \$0.55/unit.

# Sensitivity Ranges: Range of Optimality

## Stability of Production Quantities Under Margin Fluctuations

### Par, Inc. Profit Ranges ($c_j$)

| Product | Base $c_j$ | Valid Range |
|---|---|---|
| Standard ($x_1$) | \$10.00 | **[\$6.50, \$13.50]** |
| Deluxe ($x_2$) | \$9.00 | **[\$6.67, \$14.29]** |

- Optimal mix $(540, 252)$ remains **100% invariant**.
- Profit adjusts linearly:

$$\Delta Z^* = x_j^* \times \Delta c_j$$

### Managerial Interpretation

- **Corner Point Invariance:** As long as unit profits stay within these ranges, production quantities $(540, 252)$ **do not change**.
- **Operational Stability:** Routine price variations inside this band require zero factory re-tooling.

# Sensitivity Ranges: Range of Feasibility

## Valid Intervals for Constant Marginal Capacity Values

### Par, Inc. Capacity Ranges ($b_i$)

| Department | RHS | Allowable Range |
|---|---|---|
| Cutting ($\pi = \$4.38$) | 630 hr | **[495.6, 682.4]** |
| Finishing ($\pi = \$6.94$) | 708 hr | **[580.0, 900.0]** |

- Shadow price $\pi_i$ remains **strictly constant**.
- Surplus depts (Sewing, Insp.) have $\infty$ upper limit.

### Managerial Implications

- **Constant Marginal Value:** Each added hour within this range yields **exactly the shadow price**.
- **Threshold Breakpoints:** Beyond these limits, binding constraints shift, altering the optimal basis and shadow prices.

# Case Study: Parket Sisters Pencil & Pen Factory

## Multi-Product, Multi-Material Production Optimization

### Products & Profit Margins

- $X_1$: **Ballpoint Pens** ($3.00 profit)
- $X_2$: **Mechanical Pencils** ($3.00 profit)
- $X_3$: **Fountain Pens** ($5.00 profit)

### Material Inventory Limits

- **Plastic:** 1,000 oz available
- **Chrome:** 1,200 oz available
- **Stainless Steel:** 2,000 oz available

### Linear Programming Model

$$\text{Maximize } Z = 3.0X_1 + 3.0X_2 + 5.0X_3$$

**Subject to:**

$$\begin{aligned} \text{Plastic:} \quad & 1.2X_1 + 1.7X_2 + 1.2X_3 \leq 1000 \\ \text{Chrome:} \quad & 0.8X_1 + 0.0X_2 + 2.3X_3 \leq 1200 \\ \text{Steel:} \quad & 2.0X_1 + 3.0X_2 + 4.5X_3 \leq 2000 \\ & X_1, X_2, X_3 \geq 0 \end{aligned}$$

### Decisions Ahead

Extra steel offer, supply cut, and R&D redesign.

# Parket Sisters: PuLP Optimization Dashboard

Baseline Primal Production Plan and Dual Resource Audit

## Primal Production Plan (Optimal Objective: $Z^* = \$2,766.67$)

| Product | Quantity ($X_j^*$) | Profit | Reduced Cost | Status |
|---|---|---|---|---|
| $X_1$ (Ballpoint) | 700.00 units | \$3.00 | \$0.0000 | **In Production** |
| $X_2$ (Mech. Pencil) | 0.00 units | \$3.00 | -\$1.3833 | **Excluded** |
| $X_3$ (Fountain Pen) | 133.33 units | \$5.00 | \$0.0000 | **In Production** |

## Dual Resource Values & Capacity Utilization

| Material | Available | Used | Slack | Shadow Price | Status |
|---|---|---|---|---|---|
| Plastic | 1,000 oz | 1,000 oz | 0 oz | \$1.1667 / oz | **Bottleneck** |
| Chrome | 1,200 oz | 866.7 oz | 333.3 oz | \$0.0000 / oz | Surplus |
| Stainless Steel | 2,000 oz | 2,000 oz | 0 oz | \$0.8000 / oz | **Bottleneck** |

# Decision 1: Supplier Offer for Extra Steel

## Capitalizing on Bottleneck Capacity Expansion

### The Supplier's Offer

A distributor offers an additional **500 oz of Stainless Steel** at a **$0.60/oz premium** above standard contract price.

### 3-Step Evaluation

1. **Range of Feasibility:**
   - Allowable increase: 555.56 oz.
   - $500 \leq 555.56 \implies \pi_{\text{steel}} = \$\mathbf{0.80}$.
2. **Marginal Spread:**
   - Value: **$0.80/oz** vs. Cost: **$0.60/oz**
   - Net Spread: $+\$\mathbf{0.20}$/oz

### 3. Total Net Profit Impact

$$\Delta\text{Gross} = 500 \times \$0.80 = +\$400.00$$
$$\text{Cost} = 500 \times \$0.60 = -\$300.00$$
$$\mathbf{\Delta}\text{Net} = +\$\mathbf{100.00}$$

### Managerial Decision

**ACCEPT THE OFFER!**

Net company profit increases by **+$100.00**.

# Decision 2: Supplier Disruption & Chrome Shortage

## Assessing Supply Chain Shocks Using Slack Buffers

### Disruption Scenario

A logistics delay cuts Chrome supply from **1,200 oz down to 1,000 oz** (a 200 oz reduction). Should management panic?

### Slack Buffer Analysis

- **Optimal Usage:** 866.67 oz
- **Available Slack:** **333.33** oz
- **Disruption:**
  200 oz $\leq$ 333.33 oz
- Non-binding
  $\implies$ $\pi_{\text{chrome}} = \$\mathbf{0.00}$

### Impact on Operations

- **Profit Impact:** $\Delta Z^* = \$\mathbf{0.00}$
- **Production Mix:** $(700, 133.3)$ preserved
- **Remaining Slack:** **133.33** oz

### Decision: DO NOT PANIC!

Buffer absorbs 100% of shock with zero loss.

# Decision 3: R&D Redesign of Mechanical Pencil

## Pricing Out a Product Redesign Using Shadow Prices

### R&D Engineering Redesign Proposal

Original pencil is unprofitable ($d_2 = -\$1.38$). Redesigned pencil: profit remains **\$3.00**, but consumes **1.1 oz Plastic**, **2.0 oz Chrome**, and **2.0 oz Stainless Steel**.

| **Material** | **Usage** | **Opp. Cost** |
|---|---|---|
| Plastic | 1.1 oz | \$1.2833 |
| Chrome (Surplus) | 2.0 oz | \$0.0000 |
| Stainless Steel | 2.0 oz | \$1.6000 |
| **Total Hurdle Rate:** | | **\$2.8833** |

*Surplus Chrome adds \$0.00 to opportunity cost!*

### Hurdle Rate Evaluation

$$d_2^{\text{new}} = \$3.00 - \$2.88 = +\mathbf{\$0.12} > \mathbf{0}$$

### Managerial Decision

**APPROVE REDESIGN FOR PRODUCTION!**

Clears hurdle rate; increases company profit.

# Executive Summary: Sensitivity Decision Matrix

A 4-Quadrant Framework for Operational Decision-Making

## 1. Expanding Bottlenecks ($\pi_i > 0$)

- **Action:** Authorize overtime or supplier buy.
- **Pricing Rule:** Max price = Shadow Price ($\pi_i$).
- **Limit:** Valid up to Allowable Increase.

## 2. Surplus Resources ($\text{Slack}_i > 0$)

- **Action:** Never pay premiums for idle assets.
- **Pricing Rule:** Shadow price is strictly **\$0.00**.
- **Disruptions:** Absorbed free within slack.

## 3. New Products (Reduced Cost)

- **Action:** Price out scarce resource usage.
- **Hurdle Rule:** Launch if Unit Profit > Opp. Cost.
- **Costing:** Non-binding inputs cost **\$0.00**.

## 4. Price Changes ($c_j$)

- **Action:** Track market price fluctuations.
- **Rule:** Mix stays invariant within Range.
- **Profit:** Linear change $\Delta Z^* = x_j^* \Delta c_j$.